A Unified Framework for Multi-Source Respiratory Sound Classification with Data Augmentation

Qianpu Wu
High School Student
WLSA Shanghai Academy
Shanghai, China
wuqianpu@gmail.com

Jun Xiao
Network Infrastructure Department
Nokia Solutions (Shanghai) Co., Ltd
Shanghai, China
Jun.1.xiao@nokia.com

Liu Yang
Gaoling School of Artificial Intelligence
Renmin University of China
Beijing, China
yangliu_stars@163.com

***Abstract*—Respiratory diseases cause substantial morbidity and mortality worldwide. Computer-aided pulmonary auscultation provides a non-invasive and accessible approach to respiratory assessment; however, reliable automated classification remains challenging because respiratory sounds vary across recording conditions and data sources. To address this challenge, we propose a multi-source respiratory sound classification framework that constructs a unified Normal–Abnormal dataset from public and self-collected recordings. The framework applies dataset augmentation to reduce sample-size imbalance across data sources, extracts Mel-frequency cepstral coefficients (MFCCs) as acoustic features, and evaluates seven representative neural architectures under a common training protocol. Experimental results show that CNN-BiGRU achieved the highest mean accuracy of 0.7631, whereas CNN achieved the highest mean balanced accuracy of 0.7691 and attained the highest mean area under the receiver operating characteristic curve (AUROC), reaching 0.8447. Compared with the original-only setting, the expanded setting produced average gains of 4.50, 3.39, and 5.22 percentage points in accuracy, balanced accuracy, and AUROC, respectively, across the seven architectures. This work establishes a unified framework for integrating heterogeneous respiratory sound data and evaluating diverse neural architectures, providing a systematic basis for future research on automated respiratory sound classification.**

***Index Terms*—respiratory sounds, multi-source data, deep learning, data augmentation**

I. Introduction

Chronic respiratory diseases remain a major global health burden, with an estimated 569.2 million prevalent cases and 4.2 million deaths worldwide in 2023 [1]. Lung auscultation provides a simple, rapid, and non-invasive means of assessing respiratory sounds. Adventitious sounds, such as crackles, wheezes, and rhonchi, may offer clinically relevant clues to underlying respiratory abnormalities [2]. Despite its practical value, conventional lung auscultation relies on subjective interpretation and may be influenced by examiner experience, sound quality, and listening conditions [3]. Computerized respiratory sound analysis has therefore been explored as a means of supporting more consistent and reproducible assessment. In a typical pipeline, digital recordings are transformed into acoustic representations, which are then analyzed by computational models to distinguish normal from abnormal respiratory sounds. Recent studies have investigated a range of neural architectures for this task. Acharya et al. combined convolutional and recurrent networks with Mel-spectrogram inputs to jointly model spectral and temporal information [4]. To address limited training data, RespireNet integrated a CNN with device-specific fine-tuning, concatenation-based augmentation, blank-region clipping, and smart padding [5]. Bae et al. later combined a pretrained Audio Spectrogram Transformer (AST) with Patch-Mix contrastive learning, highlighting the potential of pretrained attention-based models for respiratory sound classification [6]. However, advances in model architecture do not by themselves address the challenges arising from the data. Respiratory sound datasets differ in acquisition devices, participant populations, annotation protocols, and sample definitions, which can lead to source-dependent distribution differences. Hsu et al. reported limited transferability between lung- and tracheal-sound domains and showed that mixed-set training or domain adaptation improved performance on several respiratory event-detection tasks [7]. Kim et al. treated recordings acquired with different electronic stethoscopes as separate domains and applied cross-domain adaptation to reduce device-related dependence [8]. Lungmix evaluated source-to-source transfer across ICBHI, SPRSound, and HF, observing substantial performance degradation on unseen datasets and relating it to inconsistencies in data collection and annotation [9]. Collectively, these studies suggest that data-source composition and heterogeneity should be considered explicit experimental factors alongside model architecture.

Motivated by these observations, this study presents a data-centric framework for multi-source respiratory sound classification. The framework integrates respiratory-cycle samples from ICBHI 2017 [10], event-level samples from SPRSound [11], and fixed-duration windows from a self-collected dataset into a unified binary normal–abnormal classification task. To reduce source-level sample-size imbalance, the framework applies deterministic offline gain perturbation, non-circular time shifting, and additive Gaussian noise for source-balancing dataset expansion. All audio samples are then resampled and represented using Mel-frequency cepstral coefficient (MFCC) features. Under the same feature representation and training protocol, seven representative architectures covering recurrent, convolutional, hybrid CNN–RNN, lightweight convolutional, and Transformer-based models are systematically evaluated.

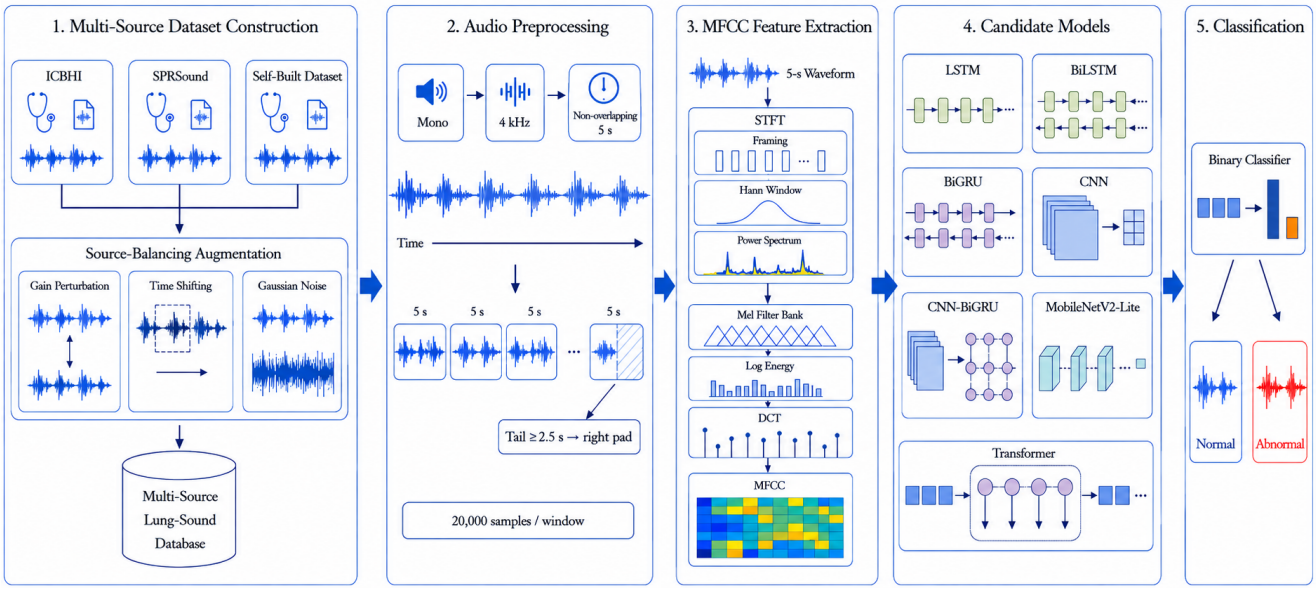


Fig. 1. Overview of the multi-source respiratory sound classification framework.

II. Methodology

A. Overall Framework

As illustrated in Fig. 1, the framework comprises five stages: (1) multi-source dataset construction with source-balancing expansion, (2) unified audio preprocessing, (3) MFCC feature extraction, (4) evaluation of candidate neural architectures, and (5) sample-level binary classification. Respiratory-cycle samples from ICBHI 2017, annotated sound events from SPRSound, and fixed-duration windows derived from self-collected recordings are organized under a shared Normal–Abnormal label space. The underrepresented self-collected source is expanded offline to reduce source-size imbalance. All samples are subsequently converted to a common 4-kHz, 5-s waveform format and represented using MFCC-16 features. Seven representative neural architectures are then evaluated using the same input representation, producing sample-level Normal or Abnormal predictions.

B. Multi-Source Dataset Construction

To construct the multi-source respiratory sound dataset, we combine ICBHI 2017, SPRSound, and self-collected respiratory sound recordings. Samples from all three sources are mapped to a common Normal–Abnormal label space.

ICBHI 2017. ICBHI 2017 provides officially annotated respiratory cycles [10], with each cycle treated as one sample. Cycles containing neither crackles nor wheezes are labeled Normal, whereas those containing crackles, wheezes, or both are labeled Abnormal.

SPRSound. SPRSound provides annotated respiratory sound events [11], with each valid event treated as one sample. Before sample selection, 187 recordings officially marked as Poor Quality are excluded. Events labeled Normal retain the Normal label, whereas Rhonchi, Wheeze, Stridor, Coarse Crackle, Fine Crackle, and Wheeze+Crackle events are mapped to Abnormal.

Self-collected respiratory sound recordings. The third source consists of respiratory sound recordings collected from volunteers recruited among relatives and acquaintances of the research team. Recordings were obtained in a quiet environment using an ETZ-2 electronic stethoscope connected to a smartphone, with participants in either a supine or upright seated position. The recordings were acquired as single-channel audio at a sampling rate of 44.1 kHz. Each participant was assigned a de-identified code, and basic metadata, including age and sex, were recorded. The self-collected source comprises 77 recordings from 11 participants. For each participant, recordings were obtained from seven auscultation sites: the trachea and the bilateral anterior, lateral, and posterior chest regions. Each recording was segmented from its beginning into consecutive, non-overlapping 5-s windows. The final shorter window was retained and padded during preprocessing. This procedure yielded 459 samples, all assigned the Normal label according to the participant recruitment and labeling criteria adopted in this study.

Overall, the original multi-source dataset contains 16,446 respiratory sound samples, including 10,988 Normal and 5,458 Abnormal samples, as summarized in Table I.

C. Data Augmentation

The three data sources differ substantially in sample size. To alleviate this source-size imbalance, the original dataset is first divided by subject into fixed training, validation, and test partitions, after which offline waveform augmentation is applied exclusively to the self-collected samples in the training partition. No augmented samples are introduced into the validation or test partitions. Three transformations are employed: gain perturbation, non-circular time shifting, and additive

TABLE I
ORIGINAL MULTI-SOURCE DATASET

Source	Sample unit	Normal	Abnormal	Total
ICBHI 2017	Respiratory cycle	3,642	3,256	6,898
SPRSound	Annotated sound event	6,887	2,202	9,089
Self-collected	Non-overlapping 5-s window	459	0	459
Total	–	**10,988**	**5,458**	**16,446**

Gaussian noise. These transformations introduce variations in signal amplitude, temporal position, and noise characteristics, respectively. Here, source balancing refers to reducing disparities in sample counts among acquisition sources rather than balancing the Normal and Abnormal classes. Because the self-collected source contains only Normal samples, the expansion increases the representation of that acquisition source but does not constitute class balancing.

Gain perturbation. Given an original waveform $x[n]$, the augmented waveform is

$$x_{\text{gain}}[n] = \alpha x[n], \tag{1}$$

where $\alpha > 0$ denotes the gain factor and is sampled uniformly from $[0.8, 1.2]$. A value of $\alpha > 1$ amplifies the waveform, whereas $0 < \alpha < 1$ attenuates it. This operation changes the overall signal amplitude while preserving the temporal arrangement of the respiratory sound patterns.

Non-circular time shifting. For a shift of τ samples, the transformed waveform is

$$x_{\text{shift}}[n] = \begin{cases} x[n-\tau], & 0 \le n-\tau < N, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where N denotes the waveform length, and τ is sampled uniformly from the nonzero integers in $[-2{,}000, 2{,}000]$, corresponding to at most ± 0.5 s at 4 kHz. Unlike circular shifting, samples moved beyond the waveform boundary are discarded instead of being wrapped to the opposite end. The newly exposed region is filled with zeros. This operation changes the position of respiratory sound patterns within the input segment while preserving their internal temporal order.

Additive Gaussian noise. The augmented waveform is

$$x_{\text{noise}}[n] = x[n] + \epsilon[n], \qquad \epsilon[n] \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

where $\epsilon[n]$ denotes zero-mean Gaussian noise and σ controls its standard deviation. The SNR is sampled uniformly from $[20, 35]$ dB, with the noise scaled to a target RMS of $\text{RMS}(x) \times 10^{-\text{SNR}/20}$. This transformation increases waveform variability by superimposing random perturbations on the original respiratory sound signal.

Together, the three transformations introduce complementary variations in amplitude, temporal position, and noise characteristics. Fig. 2 compares the time-domain waveform of the same respiratory sound sample before and after each augmentation operation. The training-only expansion generates 2,352 additional samples. Consequently, the total number of original and augmented samples increases from 16,446 in the original-only setting to 18,798 in the expanded setting, while the fixed validation and test partitions remain unchanged.

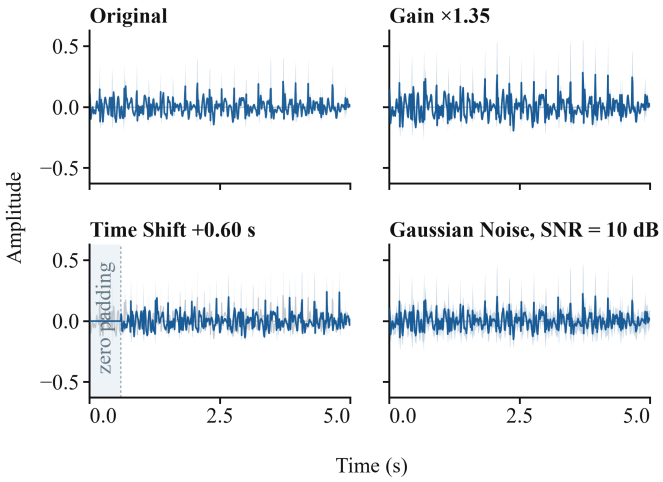


Fig. 2. Time-domain waveform augmentation examples.

D. Preprocessing and Feature Extraction

To ensure a consistent input format across the three data sources, all respiratory sound signals are converted to mono, resampled to 4 kHz, and adjusted to a fixed duration of 5 s. Signals shorter than 5 s are left-padded with zeros, whereas those longer than 5 s are truncated on the right. Mel-frequency cepstral coefficients (MFCCs) are then extracted to provide a compact representation of the short-term spectral characteristics of respiratory sounds [12]. Each waveform is first transformed into a short-time power spectrum. The spectrum is subsequently projected onto the Mel frequency scale using a Mel filter bank, followed by logarithmic compression of the filter-bank energies. Finally, a discrete cosine transform (DCT) is applied to the log-Mel energies to obtain the MFCCs. The m-th MFCC coefficient is

$$c_m = \gamma_m \sum_{k=1}^{K} \log(E_k) \cos\left[\frac{\pi m}{K}\left(k - \frac{1}{2}\right)\right], \tag{4}$$

where E_k denotes the energy of the k-th Mel filter, K is the number of Mel filters, γ_m is the DCT normalization factor, and c_m is the resulting MFCC coefficient. Sixteen coefficients are retained to form the MFCC-16 representation. MFCC extraction uses an FFT length of 1,024 samples and a hop length of 512 samples. Under this configuration, each respiratory sound sample is represented by a fixed 40×16 feature matrix, comprising 40 temporal frames and 16 MFCC coefficients per frame. The same MFCC-16 representation is used throughout all experiments, providing a consistent acoustic input for subsequent model comparisons.

E. Candidate Neural Architectures

To examine how different neural modeling mechanisms process the same respiratory sound representation, we select seven

TABLE II
MODEL CONFIGURATIONS AND PARAMETERS

Model	Architecture	Parameters
LSTM	2-layer LSTM; hidden size 64	54,402
BiLSTM	2-layer bidirectional LSTM; hidden size 64 per direction	141,570
CNN	Three Conv2D blocks; channels 16–32–64	23,538
BiGRU	2-layer bidirectional GRU; hidden size 64 per direction	106,242
CNN-BiGRU	Two Conv1D layers with 32 channels; 1-layer BiGRU with hidden size 64 per direction	42,626
MobileNetV2	Inverted-residual stages with 16–24–32–64 channels; final 128 channels	138,338
Transformer	Two encoder layers; four attention heads; embedding size 64; feed-forward size 128	70,850

candidate architectures from five model families. The recurrent group comprises long short-term memory (LSTM) [13], bidirectional LSTM (BiLSTM), and bidirectional gated recurrent unit (BiGRU) models [14]. LSTM serves as a unidirectional temporal baseline, whereas BiLSTM and BiGRU incorporate contextual information from both temporal directions. Compared with LSTM, GRU uses fewer gating components while retaining recurrent sequence-modeling capability. The convolutional group is represented by CNN, which extracts local patterns from the MFCC time–coefficient map without explicit recurrent operations. Hybrid convolutional–recurrent modeling is represented by CNN-BiGRU, which first extracts local features using convolutional layers and subsequently models their bidirectional temporal dependencies. MobileNetV2 provides a lightweight convolutional comparison through inverted residual blocks and depthwise separable convolutions [15]. Finally, Compact Transformer represents attention-based modeling by applying self-attention [16] to the 40-frame MFCC sequence, allowing us to examine its suitability for short, low-dimensional respiratory sound representations.

Table II summarizes the model configurations used in the experiments and their corresponding numbers of trainable parameters. For bidirectional recurrent models, the reported hidden size refers to each temporal direction. Each architecture ends with a binary classification head that predicts Normal or Abnormal respiratory sounds.

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

The original dataset is partitioned at the subject level within each source, targeting training, validation, and test proportions of 70%, 10%, and 20%, respectively. The original-only and expanded settings share the same validation and test partitions; the expanded setting differs only by adding augmented self-collected samples to the training partition. All models are trained using unweighted cross-entropy loss and the Adam optimizer, with a learning rate of 0.001, a weight decay of 0.0001, a batch size of 128, and a maximum of 500 epochs. Each model–setting combination is repeated three times, and the results are reported as mean ± standard deviation. The evaluation metrics include Accuracy, Sensitivity, Specificity, Balanced Accuracy, Precision, F1, and AUROC, with Abnormal treated as the positive class. Accuracy, Balanced Accuracy, and AUROC are used to characterize overall performance, while Sensitivity, Specificity, Precision, and F1 provide complementary information about class-specific behavior.

B. Results and Analysis

Table III summarizes the test performance of the candidate neural architectures under the expanded training setting. Results are reported as mean ± standard deviation over three runs, with the highest mean value for each metric highlighted in bold. CNN-BiGRU achieves the highest mean Accuracy (0.7631), together with the highest abnormal-class Precision. CNN obtains the highest mean Balanced Accuracy (0.7691) and AUROC (0.8447), together with the highest Sensitivity and abnormal-class F1. CNN-BiGRU achieves the second-highest Balanced Accuracy (0.7496), followed by BiLSTM with a Balanced Accuracy of 0.7396. The highest Balanced Accuracy is obtained by CNN, followed by CNN-BiGRU and BiLSTM. CNN-BiGRU further combines local convolutional processing with bidirectional temporal aggregation, which may contribute to its strong discrimination across decision thresholds. The architectures also exhibit different class-level trade-offs. CNN achieves the highest mean Sensitivity (0.8587) but a relatively low Specificity (0.6795), indicating a greater

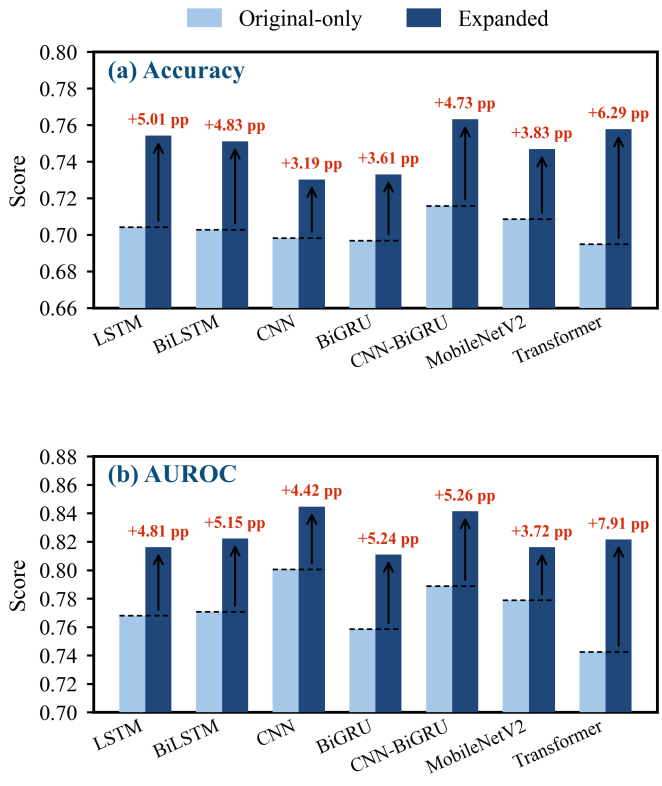


Fig. 3. Mean Accuracy and AUROC under the original-only and expanded training settings. Red annotations indicate expanded-minus-original changes in percentage points.

TABLE III
MODEL PERFORMANCE UNDER THE EXPANDED TRAINING SETTING

Model	Accuracy	Sensitivity	Specificity	Bal. Acc.	Abn. Precision	Abn. F1	AUROC
LSTM	0.7543 ± 0.0137	0.6039 ± 0.0630	$\mathbf{0.8136 \pm 0.0223}$	0.7088 ± 0.0253	0.5613 ± 0.0216	0.5808 ± 0.0354	0.8161 ± 0.0170
BiLSTM	0.7511 ± 0.0200	0.7130 ± 0.0284	0.7662 ± 0.0282	0.7396 ± 0.0176	0.5471 ± 0.0286	0.6187 ± 0.0227	0.8222 ± 0.0167
CNN	0.7302 ± 0.0175	$\mathbf{0.8587 \pm 0.0449}$	0.6795 ± 0.0367	$\mathbf{0.7691 \pm 0.0130}$	0.5148 ± 0.0194	$\mathbf{0.6431 \pm 0.0133}$	$\mathbf{0.8447 \pm 0.0174}$
BiGRU	0.7330 ± 0.0213	0.7086 ± 0.0898	0.7427 ± 0.0651	0.7256 ± 0.0125	0.5249 ± 0.0326	0.5995 ± 0.0120	0.8110 ± 0.0101
CNN-BiGRU	$\mathbf{0.7631 \pm 0.0089}$	0.7183 ± 0.0626	0.7809 ± 0.0360	0.7496 ± 0.0139	$\mathbf{0.5655 \pm 0.0191}$	0.6313 ± 0.0139	0.8414 ± 0.0019
MobileNetV2	0.7469 ± 0.0023	0.6658 ± 0.0414	0.7789 ± 0.0155	0.7223 ± 0.0132	0.5431 ± 0.0036	0.5978 ± 0.0160	0.8160 ± 0.0095
Compact Transformer	0.7578 ± 0.0121	0.6576 ± 0.0475	0.7973 ± 0.0293	0.7274 ± 0.0139	0.5626 ± 0.0221	0.6054 ± 0.0177	0.8216 ± 0.0091

tendency toward Abnormal predictions. LSTM shows the opposite pattern, with relatively high Specificity (0.8136) but the lowest Sensitivity (0.6039). These findings are specific to the present data, feature representation, and experimental configuration rather than general rankings of the architecture families.

Fig. 3 compares the mean Accuracy and AUROC obtained under the original-only and expanded settings. Both metrics are higher under the expanded setting for every architecture, although the magnitude of the change varies across models. Compact Transformer exhibits the largest increases in Accuracy and AUROC, by 6.29 and 7.91 percentage points, respectively. However, these larger gains do not correspond to the highest absolute performance under the expanded setting: CNN-BiGRU achieves the highest Accuracy, whereas CNN obtains the highest AUROC.

IV. CONCLUSION

This study presents a multi-source respiratory sound classification framework that integrates ICBHI 2017, SPRSound, and respiratory sound recordings collected in this study into a unified Normal–Abnormal classification task. To improve the representation of the underrepresented study-collected source, gain perturbation, non-circular time shifting, and additive Gaussian noise are used for offline data augmentation. Using a common MFCC-16 representation and training protocol, seven representative neural architectures are compared. Under the expanded training setting, CNN-BiGRU achieves the highest mean Accuracy (0.7631), whereas CNN obtains the highest mean Balanced Accuracy (0.7691) and AUROC (0.8447). Compared with the original-only setting, all evaluated architectures achieve higher mean Accuracy and AUROC after dataset expansion. These findings demonstrate that both training-data composition and neural architecture selection influence multi-source respiratory sound classification, and provide a practical reference for constructing respiratory sound datasets and selecting appropriate model families. Overall, this work establishes a unified framework for integrating heterogeneous respiratory sound data and evaluating diverse neural architectures, providing a systematic basis for future research on automated respiratory sound classification.

REFERENCES

[1] J. Oh *et al.*, "Global, regional, and national burden of chronic respiratory diseases and impact of the COVID-19 pandemic, 1990–2023: A global burden of disease study," *Nature Medicine*, vol. 32, pp. 197–223, 2026.

[2] A. Bohadana, G. Izbicki, and S. S. Kraman, "Fundamentals of lung auscultation," *New England Journal of Medicine*, vol. 370, no. 8, pp. 744–751, 2014.

[3] R. X. A. Pramono, S. Bowyer, and E. Rodriguez-Villegas, "Automatic adventitious respiratory sound analysis: A systematic review," *PLoS ONE*, vol. 12, no. 5, p. e0177926, 2017.

[4] J. Acharya and A. Basu, "Deep neural network for respiratory sound classification in wearable devices enabled by patient specific model tuning," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 14, no. 3, pp. 535–544, 2020.

[5] S. Gairola, F. Tom, N. Kwatra, and M. Jain, "RespireNet: A deep neural network for accurately detecting abnormal lung sounds in limited data setting," in *Proceedings of the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2021, pp. 527–530.

[6] S. Bae *et al.*, "Patch-mix contrastive learning with audio spectrogram transformer on respiratory sound classification," in *Proceedings of Interspeech 2023*, 2023, pp. 5436–5440.

[7] F.-S. Hsu *et al.*, "A dual-purpose deep learning model for auscultated lung and tracheal sound analysis based on mixed set training," *Biomedical Signal Processing and Control*, vol. 86, p. 105222, 2023.

[8] J.-W. Kim, S. Bae, W.-Y. Cho, B. Lee, and H.-Y. Jung, "Stethoscope-guided supervised contrastive learning for cross-domain adaptation on respiratory sound classification," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 1431–1435.

[9] S. Ge, W. Zhang, S. Xie, B. Yan, and Z. Wang, "Lungmix: A mixup-based strategy for generalization in respiratory sound classification," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.

[10] B. M. Rocha *et al.*, "An open access database for the evaluation of respiratory sound classification algorithms," *Physiological Measurement*, vol. 40, no. 3, p. 035001, 2019.

[11] Q. Zhang *et al.*, "SPRSound: Open-source SJTU paediatric respiratory sound database," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 16, no. 5, pp. 867–881, 2022.

[12] W. Zhou, L. Yu, M. Zhang, and W. Xiao, "A low power respiratory sound diagnosis processing unit based on LSTM for wearable health monitoring," *Biomedical Engineering / Biomedizinische Technik*, vol. 68, no. 5, pp. 469–480, 2023.

[13] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[14] K. Cho *et al.*, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734.

[15] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510–4520.

[16] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 2017, pp. 5998–6008.